FlyDreamAir Booking System

Group ID: T11

Group Members

For team contribution, Zac worked on project coordination, the introduction, business case, meeting records, and UI design. Jay worked on the WBS, schedule, milestone reports, and conclusion. Nikita completed the risk management, testing, project closing, and lessons learned. Zexi completed the project selection justification, project charter, scope statement, and requirements. Cheung worked on cost estimation, version control evidence, and functionality implementation

Member	Main Contribution
Zac Wastie	Project coordination, introduction, business case, meeting records, and UI design
Jay Segar	WBS, WBS dictionary, project schedule, milestone reports, and conclusion
Nikita Kravchenko	Risk management, testing, project closing, and lessons learned
Zexi Dong	Project selection justification, project charter, project scope statement and requirements
Sui San Cheung	Cost estimation, version control evidence, and functionality implementation

Project Overview & Team Contribution

OBJECTIVE

Develop a working prototype that allows customers to search flights, book tickets, select seats, purchase in-flight services and manage reservations.

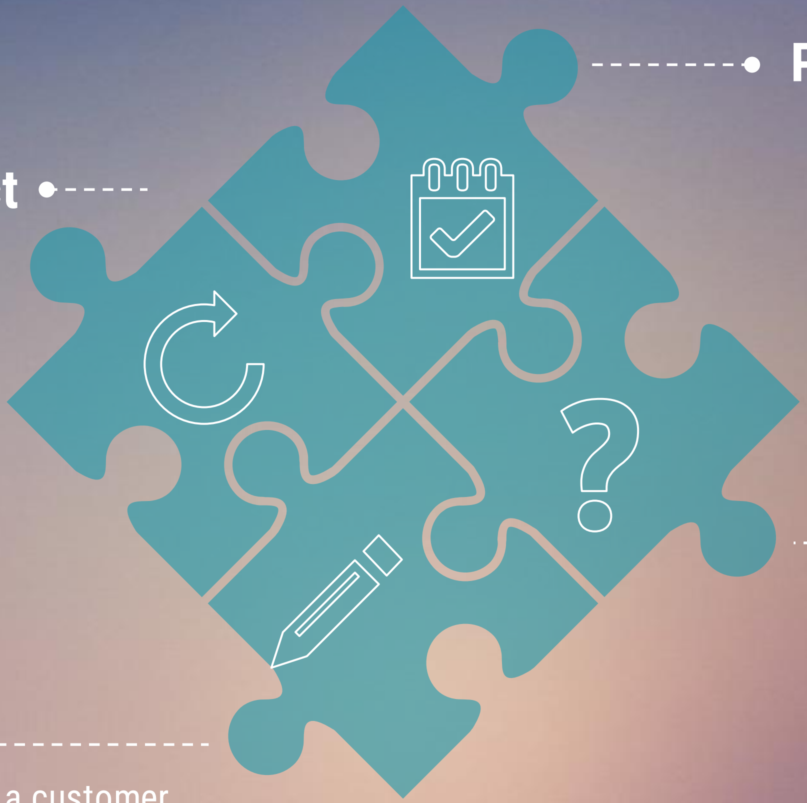
DELIVERABLES

- Prototype system
- Project report
- Meeting records
- Milestone reports
- Testing evidence
- Final presentation and demo

CONSTRAINTS

Selected Project

Project 1 –
Customer booking and reservation system



Planned Timeframe

2 years

Budget

$1 million

Objective

Develop a prototype of a customer reservation system

Project Selection & Business Case

Business Background

FlyDreamAir is a major airline operating domestic and international flights. The company wants to digitalise its business processes.

Search flights

Current Problems

The existing booking and customer management process is outdated or not fully integrated, causing slower booking, poor usability and booking management difficulty.

Manage bookings

Project Need

A modern customer booking platform is needed to improve customer experience and internal efficiency through one digital system.

Improve experience

Project Selection & Business Case

SELECTION METHOD

Used weighted scoring model from course content

Compared Project 1, Project 2 and Project 3

Criteria: business alignment, customer impact, technology realism, timeframe feasibility and risk

WHY PROJECT 1 WON

Project 1 directly supports FlyDreamAir's digital transformation and affects a wider customer group than loyalty or lounge systems.

BUSINESS CASE SUMMARY

Problem: inefficient and fragmented booking process

Opportunity: faster bookings and easier reservation management

Benefits: better customer satisfaction, fewer errors and improved operations

Feasible because prototype uses free tools, basic web technology and simulated data

Our Approach to the Problem

Select Project 1

We selected Project 1 because it had the strongest business value and customer impact.

Define Scope

We focused on flight search, booking, seat selection, in-flight services, and reservation management.

Manage the Project

We used project charter, WBS, schedule, risk management, cost estimation, and weekly meetings.

Build & Test

We developed a working prototype and tested core functions such as login, flight search, booking, seat selection, and cancellation.

The report uses project management artefacts such as the charter, WBS, schedule, risk register, cost estimation, milestone tracking and meeting records. The prototype was then refined through team meetings and tutor feedback.

Project Scope

In Scope

Login / registration
Flight search
Ticket booking
Seat selection
In-flight services
Booking confirmation
Manage / cancel reservations

Out of Scope

Real payment gateway
Real airline database
Live flight tracking
Loyalty program
Lounge management
Full admin analytics

This scope protects the team from scope creep while still showing the complete reservation workflow.

Key Requirements & Scope Control

FUNCTIONAL

Create account and log in

Search available flights

Select a flight and enter passenger details

Choose seats and add food/drinks

Generate confirmation

View, manage or cancel bookings

NON-FUNCTIONAL

Clear and simple UI

Consistent page design

Reasonable loading time with sample data

Accurate storage in prototype environment

Basic login security

Reliable enough for demo

CHANGE CONTROL

Requirements were reviewed through team discussions and tutor feedback. Complex functions such as real payments, live flight data and admin analytics were excluded to control scope creep.

Scope stays achievable

Project Management Deliverables

PROJECT CHARTER

Approved in Week 4 after selecting Project 1. Defines manager, objective, budget, timeframe and success criteria.

Week 4

WBS & DICTIONARY

Breaks work into initiation, requirements, design, development, testing, documentation and presentation tasks.

Week 5

PROJECT SCHEDULE

Uses proposed and actual Gantt charts to plan milestones and compare progress over the semester.

Week 6

MILESTONE REPORTS

Tracks planned outcomes, actual progress, problems found and next steps for each milestone.

Week 7

Cost Estimation

FUNCTION POINT COUNT

Unadjusted Function Points: 72 UFP

Total Degree of Influence: 30

VAF = 0.65 + (0.01 × 30) = 0.95

Adjusted Function Points: 68.4 AFP

EFFORT

68.4 AFP × 8 hours =

547.2 person-hours

SUMMARY OF COSTS

Labour: $27,360

Tools/licences: $0

Computing resources: $0

Contingency 15%: $4,140

Total estimate: $31,500

The estimate is well within the $1 million notional budget because the prototype uses simulated data and free tools.

Project Execution & Tracking Evidence

PROJECT EXECUTION

1. Weekly meetings in person and online Discord used for communication

2. Tasks assigned and reviewed regularly

3. Prototype and report progress checked during meetings

TRACKING EVIDENCE

Milestone reports

Meeting records

Progress review against schedule

Proposed vs actual Gantt chart

VERSION CONTROL

GitHub repository used

Feature branches for major features

Pull requests and reviews

Commit history used as contribution evidence

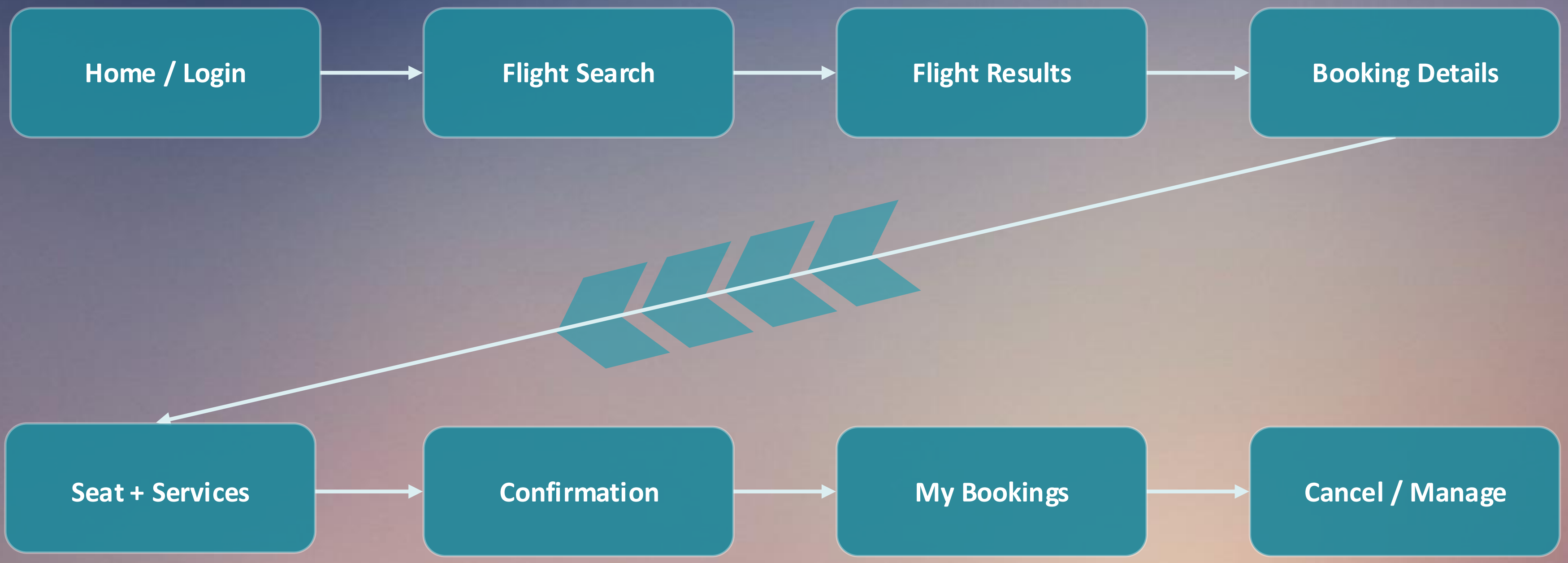
Tags for milestone releases

Risk Management

Team disorganisation	Low / Medium	Resolve issues quickly through Discord
Team miscommunication	Low / Medium	Weekly meetings and clear task clarification
Team management issues	Medium / High	Clear deadlines and task allocation
Technical issues	Medium / Medium	Team collaboration and early testing
Scope too large	Medium / High	Limit prototype to core booking features

The risks were monitored during the task using weekly meetings and direct communication.

Prototype Workflow



Demo order: start from login, search a flight, book it, choose a seat, add services, confirm the booking, then view and cancel it from My Bookings.

Functionality Implementation

FULLY IMPLEMENTED

User registration and login

Flight search and results display

Passenger information entry

Seat selection

In-flight service purchase

Booking confirmation

View existing bookings

Cancel a reservation

NOT IMPLEMENTED / MOCKED

Payment gateway: mocked as “Pay on check-in”

Real-time flight data: static hard-coded flights

Advanced admin panel: out of scope

DEMO

Project Closing & Lessons Learned

CLOSING STATUS

Prototype delivered

Project documentation completed

Meeting records maintained

Testing evidence provided

Core booking workflow demonstrated

WHAT WENT WELL

Strong collaboration

Early risk identification

Clear task allocation

Scope controlled by excluding complex functions

IMPROVEMENTS

Start testing earlier

Use a clearer progress checklist

Maintain more consistent communication during busy weeks

Overall, the project met its intended scope and gave the team practical experience in planning, documentation, risk management, prototype development and teamwork.

THANKS FOR YOUR ATTENTION